

RUPERTO PEDRO BONET CHAPLE

Email: ruperto.bonet@modelphysmat.com | Phone: +447553049140

GitHub: <https://github.com/sednabcn> | LinkedIn: <https://www.linkedin.com/in/ruperto-p-bonet-chaple-8a26651b/>

Summary

Highly experienced mathematics educator and researcher with a Ph.D. in Computational Mechanics. Over 40 years of teaching combined with recent advancements in Python development, machine learning, and real-world data science projects.

Employment History

Dec 2022 – Present

Maths Tutor | Madison Reign Agency (Kingsdale Foundation School) (Part-Time)

- Teach GCSE students with learning difficulties in a face-to-face setting.
- Develop simplified tutorial materials for easy assimilation.
- Support and challenge students to meet academic targets.

April 2021 – Dec 2022

Maths Tutor | Self-Employed (Part-Time)

- Delivered GCSE, A-Level, and university-level math tutoring online and in person.
- Prepared students for exams and assessed their performance.
- Provided exam invigilation services for various academies.

Oct 2020 – April 2021

Online Tutor (MSc. Data Science) | Worldwide University of London (Part-Time)

- Taught Calculus, Geometry, and Statistics through webinars.
- Designed lectures, marked exams, and ensured high student achievement.

Sept 2015 – Oct 2016

Associate Professor | Technical University of Babahoyo (UTB-Ecuador) (Full-Time)

- Taught undergraduate and postgraduate courses in Calculus, Statistics, and Matlab.
- Managed statistical projects in agriculture, education, and business sectors.
- Published research on shallow water equations and computational methods.

Feb 2014 – Sept 2015

Writer/Author | Funiber Editorial, Barcelona, Spain (Full-Time)

- Authored the book *Calculus and Differential* (ISBN 978-84-9079-205-6).
- Reviewed educational materials in finance mathematics.

Aug 2010 – Feb 2014

Associate Professor | National University of San Luis, Argentina (Full-Time)

- Taught engineering and mathematics courses, including Partial Differential Equations and Fourier Analysis.
 - Supervised thesis projects and conducted research on numerical methods.
-

Education

Ph.D. in Computational Mechanics

CIMEC-UNL, Argentina (Feb 1995 – Jul 1999)

Post-Doctoral Fellowship in Numerical Models for Coastal Hydrodynamics

CIMEC-UNL, Argentina (Apr 2000 – Jul 2002)

Certifications:

- Python for Data Science and AI – IBM (93.25%, 25 hrs, Jun 2019)
 - Applied AI with DL Advanced – IBM (96.88%, 25 hrs, Apr 2019)
 - Practical Time Series Analysis – State University of New York (97.03%, 25 hrs, Oct 2020)
 - Practical Reinforcement Learning – HSE University (99.13%, 26 hrs, Feb 2021)
-

Technical Skills

- **Programming Languages:** Python, Matlab, Maple, Mathematica, C++.
 - **Data Science Tools:** Jupyter Notebooks, Google Cloud Platform, Apache Spark.
 - **Machine Learning Frameworks:** TensorFlow, Scikit-learn, Keras.
 - **Statistical Analysis & Visualization:** R, Pandas, Matplotlib, Seaborn.
-

Publications

- *Mathematics I* (2016) ISBN 978-84-9079-205-6.
 - *Refraction and Diffraction of Water Waves Using Finite Elements with a DNL Boundary Condition*. Ocean Engineering Journal, Vol. 63, pp. 77-89 (2013).
 - *Numerical Stabilization of Convection-Diffusion-Reaction Problems*. Delft (2006).
 - Additional publications available upon request.
-

Conferences

- *Improved Discrete Non-Local Absorbing Boundary Condition for Helmholtz Equation*. III MACI 2011, White Bay, Argentina.

- *Outgoing Boundary Conditions for Elliptic Harbor Wave Models*. VII International Conference, Barcelona, Spain (2003).
 - *An Extended Elliptic Water Waves Model to Simulate Waves in Harbors*. Marine 2005, Oslo, Norway.
-

References

Sergio Idelshon

Ph.D. Thesis Supervisor

Email: sergio@cimne.upc.edu | Phone: +34-93-401-18-29

Aziza Sentisi

Colleague

Email: asenti01@mail.bbk.ac.uk | Phone: +447527004473

(References available for contact prior to interview.)
